Documentation for Air Quality Sensor Stations.

GitHub link: https://github.com/Wason1797/aiq_coap_server

Sensors Used

SCD41: Used to measure CO2, Temperature, and Humidity

Documentation:

> SCD41
> CO2 accuracy specifications:400-1,000 ppm: ±(50 ppm + 2.5% of reading)1,001-2,000 ppm: ±(50 ppm + 3% of reading)2,001-5,000 ppm: ±(40 ppm + 5% of reading) The SCD4x is our series of miniature CO2 sensors, which builds on the photoacoustic NDIR sensing principle and
> https://sensirion.com/products/catalog/SCD41

> https://sensirion.com/media/documents/48C4B7FB/66E05452/CD_DS_SCD4x_Datasheet_D1.pdf

ENS160: Used to measure ECO2, TVOC, and AQI

Documentation:

> https://cdn-learn.adafruit.com/downloads/pdf/adafruit-ens160-mox-gas-sensor.pdf

> https://www.mouser.com/datasheet/2/1081/SC_001224_DS_1_ENS160_Datasheet_Rev_0_95-2258311.pdf?srsltid=AfmBOoogZsJs3MxfFjPJ1QwoMelhPv_1PJKGZOGhHguNEEjIN9LRbe9v

SVM41: Used to measure Temperature, Humidity, VOC Index. and NOX Index

Documentation:

https://sensirion.com/media/documents/FBD0A26B/61EA8BFA/Sensirion_Gas_Sensors_SEK-SVM4x_Technical_Description.pdf

https://sensirion.com/media/documents/33FD6951/662A593A/HT_DS_Datasheet_SHT4x.pdf

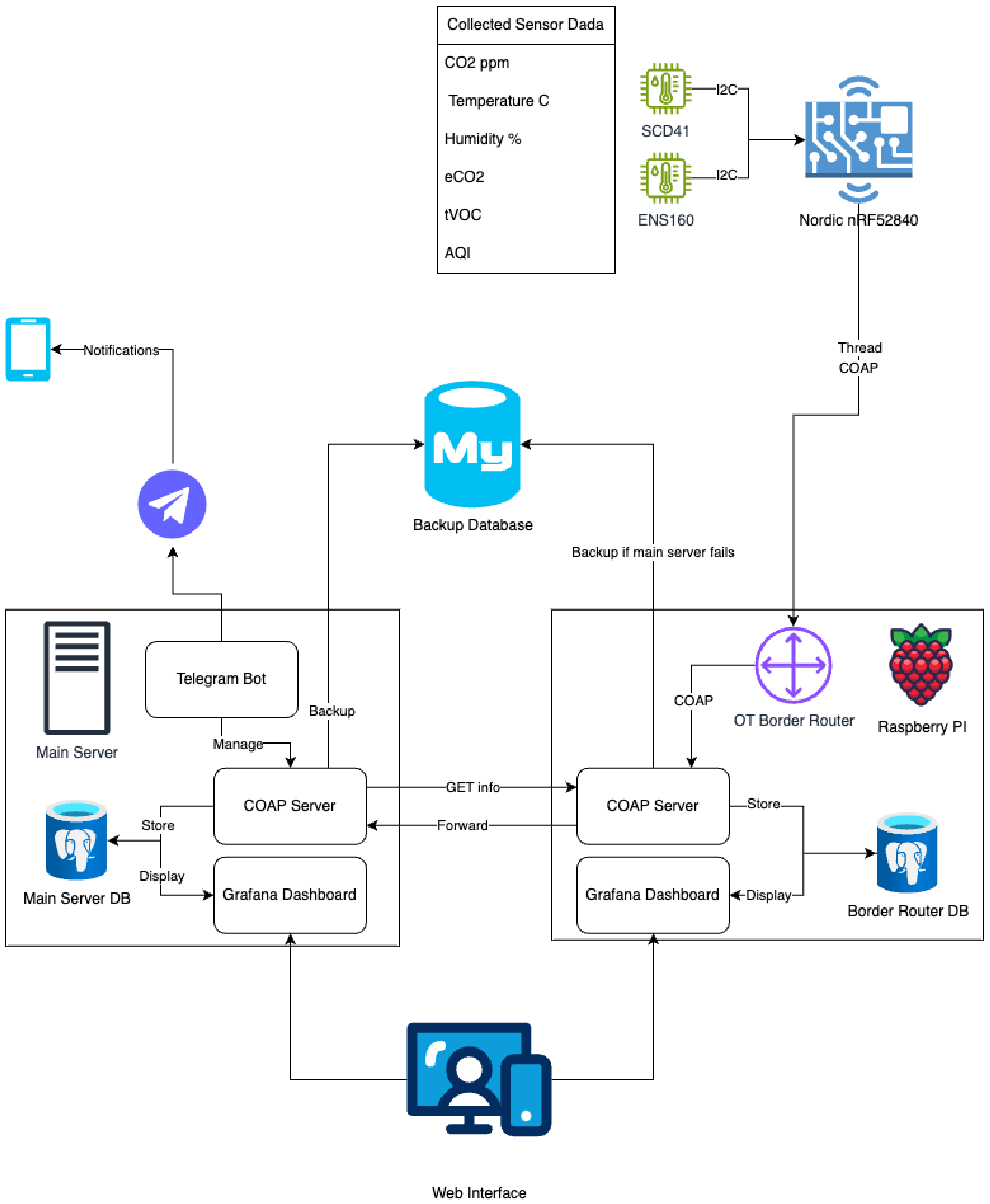
https://sensirion.com/media/documents/296373BB/6203C5DF/Sensirion_Gas_Sensors_Datasheet_SGP40.pdf

BME688: Used to measure Temperature, Humidity, Ambient Pressure, and Gas resistance

Documentation:

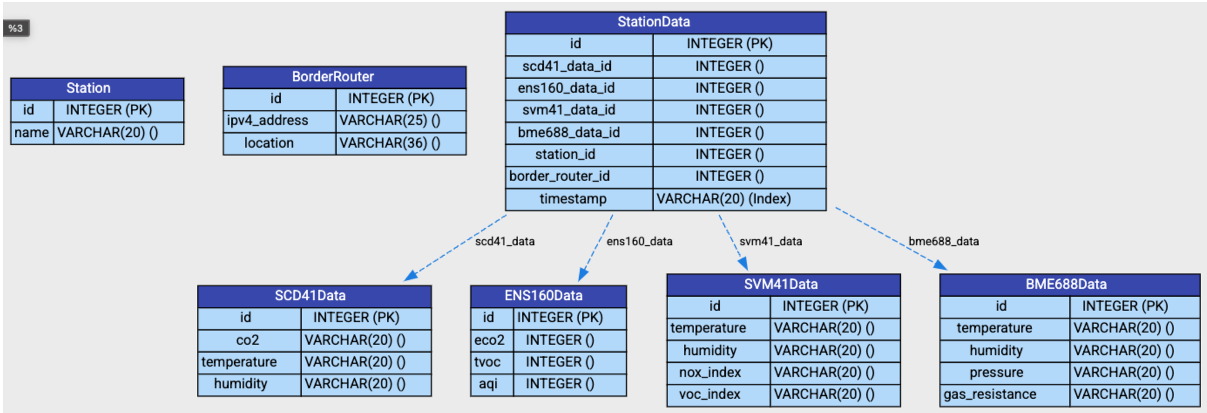
https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bme688-ds000.pdf

Architecture of the system



Note: This was the initial architecture, some upgrades are the inclusion of the SVM41, and a refactoring of the database schema to make it easier to add sensors in the future. Also due to problems with the resiliency of the raspberry pi storage, we are not using it to forward, or store data.

ERD



Telegram Bot

To use the telegram bot:

1. Write: `/start` in the chat, the bot will respond with `hello xxxxxxxx`
2. Please share with wladymir.brborich-herrera@stud.fra-uas.de the number returned by the bot, that is your telegram user id.
3. I will include the id in the allow list so you can use the bot.

Commands:

- `/data_summary` Will return the last inserted row from the database. Use this to check if the system is online and producing values
- `/station_summary` Will return a summary of registered sensor stations with id and name
- `/br_summary` Will return a summary of registered border routers